

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech Theory Examination (Nov-Dec 2017)

CL 216.01 Mathematics for civil Engineers

Date: 20.11.2017, Monday

Time: 10.00 a.m. To 01.00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper contains two sections. Each section has three questions.
2. Section I and II must be attempted in separate answer sheets. Each section carries 35 marks.
3. Figures to the right indicate marks.
4. Make suitable assumptions and draw neat figures wherever required.

SECTION – I

Q - 1 Multiple Choice Questions

[07]

- 1) The Laplace transform of e^{-7t} is _____.

(a) $\frac{1}{s+7}$ (b) $\frac{1}{s-7}$ (c) $\frac{2}{s+7}$ (d) $\frac{2}{s-7}$

- 2) An equation of the form $x^3 - 3x^2 + 2x - 3 = 0$ is called _____ equation.

(a) linear (b) quadratic (c) cubic (d) biquadratic

- 3) If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ then $\text{curl } \vec{r}$ equal to

(a) 3 (b) r (c) 0 (d) -r

- 4) Inverse Laplace transforms of $\frac{1}{(s+4)^6}$

(a) $e^{-6t} \frac{t^4}{4!}$ (b) $e^{-4t} \frac{t^6}{6!}$ (c) $e^{-4t} \frac{t^5}{5!}$ (d) $e^{-4t} \frac{t^6}{5!}$

- 5) $x^3 + 10 = 0$ has

(a) One negative root (b) one positive root (c) 3 complex roots
(d) one positive and two complex roots

- 6) Let $f(x,y,z)=c$ represent the equation of the surface. Then unit normal vector to this surface is

(a) $\frac{\text{grad } f}{|\text{grad } f|}$ (b) $\text{grad } f$ (c) $\text{div}(\text{grad } f)$ (d) $\text{curl}(\text{grad } f)$

- 7) A vector $\vec{f}$ is solenoidal if

(a) $\text{div } \vec{f} = 0$ (b) $\text{curl } \vec{f} = 0$ (c) $\text{div } \vec{f} \neq 0$ (d) $\text{curl } \vec{f} \neq 0$

Q - 2 Attempt any four out of six

[16]

- 1) Find the Laplace transform of the square wave function of period $2a$ defined as

$$f(t) = \begin{cases} k & \text{if } 0 \leq t < a \\ -k & \text{if } a \leq t < 2a \end{cases}$$

- 2) Using Laplace transform, solve the differential equation

$$\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + 5x = e^{-t} \sin t, x(0) = 0, x'(0) = 1$$

- 3) Find solution of following equation by using Cardan's Method $x^3 - 6x^2 + 6x - 5 = 0$.
- 4) Find the directional derivative of $\phi = 4xz^3 - 3x^2y^2z$ at the point (2,-1,2) along the direction normal to the surface $x^2 + y^2 + z^2 = 9$ at (1,2,2).
- 5) Prove that $\vec{F} = (y^2 \cos x + z^3)\hat{i} + (2y \sin x - 4)\hat{j} + 3xz^2\hat{k}$ is irrotational and find its scalar potential.
- 6) Find solution of following equation by using Cardon's Method

$$x^3 - 18x - 35 = 0$$

Q – 3 (a) Attempt any two out of three

[06]

- 1) Find the inverse Laplace transform of $\frac{s+29}{(s+4)(s^2+9)}$.
- 2) Find the directional derivative of $f(x, y, z) = x^2 + 3y^2 + z^2$ at the point P(2,1,3) in the direction of the vector $\hat{i} - 2\hat{k}$.
- 3) Find constants a,b,c so that $\vec{v} = (x+2y+az)\hat{i} + (bx-3y-z)\hat{j} + (4x+cy+2z)\hat{k}$ is irrotational

Q – 3 (b) Attempt any two out of three

[06]

- 1) Find the inverse Laplace transform of function $F(s) = \frac{3(s^2-1)^2}{2s^5}$.
- 2) Prove that $r^n \vec{r}$ is irrotational.
- 3) Find solution of $x^4 + 12x - 5 = 0$ by using Ferrari's Method.

SECTION – II

Q – 4 Multiple Choice Questions

[07]

- 1) A necessary and sufficient condition condition that line integral $\oint_C \vec{F} \cdot d\vec{r} = 0$ for every closed curve C is that
 (b) $\text{div } \vec{f} = 0$ (b) $\text{curl } \vec{f} = 0$ (c) $\text{div } \vec{f} \neq 0$ (d) $\text{curl } \vec{f} \neq 0$
- 2) In a tossing a fair die, the probability of getting an odd number is
 (a) 2 (b) $\frac{1}{2}$ (c) $\frac{2}{3}$ (d) $\frac{3}{4}$
- 3) The n^{th} difference of a polynomial of degree n is
 (a) constant (b) zero (c) n! (d) 2
- 4) E equal to
 (a) $1 + \Delta$ (b) $\Delta \nabla$ (c) $\Delta + \nabla$ (d) $\nabla - \Delta$
- 5) The expression $\text{curl}(\text{grad } f)$, where f is a scalar function, is
 (a) equal to $\nabla^2 f$ (b) equal to $\text{div}(\text{grad } f)$ (c) a scalar of zero magnitude
 (d) a vector of zero magnitude
- 6) The order of the difference equation $y_{n+3} - 3y_{n+1} + 2y_n = 0$ is
 (a) 1 (b) 2 (c) 3 (d) 0
- 7) The Gauss divergence theorem relates certain

- (a) surface integrals to volume integrals
- (b) surface integrals to line integrals
- (c) vector quantities to other vector quantities
- (d) line integrals to volume integrals

Q -5 Attempt any four out of six [16]

1) Evaluate $\iint \vec{F} \cdot \hat{n} \, ds$ where, $\vec{F} = (x^2 - yz)\hat{i} + (y^2 - zx)\hat{j} + (z^2 - xy)\hat{k}$ taken over the [04]

rectangular parallelepiped $x=0, y=0, z=0, x=a, y=b, z=c$.

2) An unbiased coin is tossed 6 times. Find the probability of getting (i) exactly 4 heads (ii) at least 4 heads. [04]

3) Construct polynomial for the following data: [04]

X:	4	6	8	10
Y:	1	3	8	16

Find the value of y for x=5 using Newton Interpolation formula.

4) Verify Stokes theorem for $\vec{F} = xy^2\hat{i} + y\hat{j} + z^2x\hat{k}$ for the surface of a rectangular lamina bounded by $x=0, y=0, x=1, y=2, z=0, z=1$. [04]

5) A card is selected at random from the following four cards: 10 of diamonds, 10 of spades, 8 of hearts and 6 of clubs. Let E be the event "Select a red card" and Let F be the event "Select a 10". [04]

- (a) Are E and F mutually exclusive?
- (b) Are E and F independent?

6) Show that relation between operators [04]

(i) $\nabla = 1 - E^{-1}$ (ii) $\delta = E^{\frac{1}{2}} - E^{-\frac{1}{2}}$

Q -6 (a) Attempt any two out of three [06]

1) Use divergence theorem to evaluate $\iiint_S \vec{F} \cdot d\vec{S}$, where $\vec{F} = xy^2\hat{i} + yz^2\hat{j} + zx^2\hat{k}$ and S is [03]

the surface of the sphere $x^2 + y^2 + z^2 = 1$.

2) A coin is tossed ten times. What is the probability that heads occurs six times? [03]

3) Use Lagrange's interpolation formula to find the value of y when x=10, if the values of x and y are given below: [03]

X	5	6	9	11
y	12	13	14	16

Q -6 (b) Attempt any one out of two [06]

1) There are two boxes A and B containing 4 white, 3 red and 3 white, 7 red balls respectively. A box is chosen at random and a ball is drawn from it, If the ball is white, find the probability that it is from box A. [06]

2) Verify Divergence theorem for $\vec{F} = 4xz\hat{i} - y^2\hat{j} + yz\hat{k}$ over the cube bounded by the planes $x=0, y=0, x=2, y=2, z=0, z=2$ [06]

